

SEBASTIAN MONZÓN

Full-Stack Engineer & Researcher

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RESEARCH INTERESTS

Human-Computer Interaction, Embodied AI, Computational Design & Sensing Systems

Education

University of Massachusetts Amherst

Amherst, MA

B.S. in Computer Science | CGPA: **3.68** / 4.0

09/2019 - 05/2023

- Honors Thesis from the Commonwealth Honors College
- Minor in Business

AWARDS AND HONORS

 NSF Fellow (CSGrad4US) \$159,000 covering 3 years over a 5-year fellowship period starting Fall 2026.	 Scholarships UMass Amherst Dean's Award Melvin Howard Scholarship Ruth M. White Scholarship	 HackUMass IX Hackathon Awards Won a finalist award and "Best use of Google Cloud" for a Spotify song queue voting app.
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PUBLICATIONS IN-PREPARATION

An AI-Driven Acoustic Symphony: Optimized Hydrogel Dewatering Leveraging a Diffusion-model for Music Spectrogram Inpainting

Shuvo, I., Monzón, S., Smith, G., Flores, R., Diaz-Marin, C., Gu, R., Nagle, S., Paradiso, J., Boriskin, V., & Boriskina, S.

Designed and tested an AI framework for hydrogel dewatering optimization using music generated via Stable Diffusion and soft-prompt optimization. Incorporates a Riffusion model derivative for Mel-spectrogram inpainting and CLIP latent guidance.

A Wearable Waterproof System for Long-Range Health Monitoring and Intelligent Distress Signaling

Shuvo, I., Monzón, S., & Boriskina, S.

Designed, 3D printed, and coded a wearable waterproof device with long-range radio (ESP32 w/ LoRa), IMU (MPU9250), and pulse sensor for real-time machine learning interpretation of swimmer movement and biomedical data. Programmed to display and record movement, heart rate, and heart rate variability (HRV) data in real-time with a LoRa receiver.

Musically Interpreting a Simple Generative Transformer

Cosme-Clifford, N., Symons, J., Kapoor, K., Monzón, S., & Wm. White, C.

Explored the aptitudes and failures of a GPT-style model trained on choral-style hymn MIDI files. Encoded chords into Unicode for GPT interpretation and analyzed its learned latent space for implications of AI on music.

CONFERENCES & PRESENTATIONS

2025 Material Research Society Fall Meeting, Boston

Symphony of Dewatering Water Harvesting Hydrogels using AI-Generated Music

12/2025

Will be presenting my Stable Diffusion framework for improved hydrogel water extraction at the MRS Fall conference.

Dept. of Multifunctional Metamaterials (META) at MIT

Music Generation with Stable Diffusion Inpainting

04/2025

Presented research on an AI framework I developed to improve hydrogel dewatering using Stable Diffusion inpainting methods.

ADDITIONAL RESEARCH EXPERIENCE

Department of Music, UMass Amherst

Amherst, MA

Music-Emotion Study

08/2025 - Present

Exploring human perceptual responses to music features in collaboration with my undergraduate honors thesis advisor.

- Composed several pop music pieces with fixed chord progressions across and variation across pitch, mode, and rhythmic speed.
- Collaborated with musicologists and data scientists on defining experimental test setup and survey metrics, to minimize confounding variables.
- Using EKG, EEG, and GSR to measure the physiological responses of study participants.

MAGE: Motion-to-Audio Generative autoEncoder

06/2025 - Present

A multi-modal variational auto-encoder (MMVAE) for gestural audio synthesis.

- LSTM (Long-Short Term Memory) and 1D Causal TCN (Temporal Convolutional Network) blocks for deep learning of sequential hand position data to BRAVE (Bravely Realtime Audio Variational autoEncoder) latent space mapping.
- Uses high FPS video to encode hand gestures for accurate positioning and inference of corresponding conga drum sounds.
- Created a dataset of hand-landmarks and audio encodings from a BRAVE model trained on professional conga drum recordings.
- Libraries include OpenCV, MediaPipe, and PyTorch.

Commonwealth Honors College, UMass Amherst

Amherst, MA

Towards Generative Adversarial Audio Synthesis via Latent Interpolation Between Desired Timbral Qualities

04/2023

My senior year Honors Thesis project to generative and classification models for audio.

- Researched state-of-the-art AI audio generation models including WaveGAN, WaveNet, and NSynth.
- Proposed an AI model to generate musical notes based on sound descriptors using a GAN model.
- Designed and built a multi-class classification network to categorize sounds by their instruments with 76% accuracy.
- Wrote a peer-reviewed 24-page manuscript.

INDUSTRY EXPERIENCE

Cisco Meraki

Remote

Software Engineer

06/2023 - Present

- Collaborated closely with UX designers and project managers to design, develop, and deploy scalable networking solutions for the Meraki Dashboard.
- Technical Lead for the "Power Settings" page of the Meraki Dashboard to distribute power-saving configuration options to Meraki APs.
- Resolved several time-sensitive, million dollar issues for clients across the globe by debugging Meraki's 4 million line codebase.
- Performed several interviews and onboarded software engineers in India and China.
- Designed an LLM for retrieving Slack and Confluence documents to help developers access information during the 1-week Cisco hackathon.

Beamshyft — beamshyft.com

Boston, MA

Startup CTO

2025 - Present

Low-cost housing supplies shipped directly from manufacturers

- Designed, developed, and launched a professional, dynamic website with video, catalogue, and scheduling system.
- Improved project clarity by outlining phases in a statement of work for 4 key development phases.
- Adapted development phases to meet business needs.

Cisco Meraki

Remote

Software Engineer Intern

2023

- Collaborated the UX team to develop the updated UI for the Radio Settings page using Cisco Magnetic design principles.
- Diagnosed high-priority issues to prevent dashboard outages for customers with large networks.

RELEVANT COURSES

Machine Learning (589) — Graduate level

Reinforcement Learning (H589) — Graduate level

Artificial Intelligence (383)

Uncertainty, Risk, and Decision Making (499CM)

Software Entrepreneurship (420)

Fundamentals of Music Theory (110)

ACTIVITIES

Swim-team captain · Robotics Club (computer vision) · St. Jude Children's Research Hospital fundraising · TKE recruitment chairman

SKILLS

AI & Machine Learning:

PyTorch · Tensorflow · MediaPipe · Multimodal VAE · RAVE · VRNN · Audio/Signal Processing · CLIP · Stable Diffusion

Programming & Systems: Python · Matplotlib · Librosa · Pandas · Embedded C++ · CUDA · MATLAB · React/Next.js · Ruby on Rails · SQL

Design & Fabrication: CAD (Fusion 360) · PCB Design (KiCAD) · 3D Printing

Creative Practice: Music Production (Ableton Live) · Sound Design · Adobe

Research & Interpersonal:

Problem-Solving & Critical Thinking · Creativity & Innovation (Musician · Inventor · Builder) ·

Collaboration & Communication (Research Partnerships · Recruitment Chairman) · Leadership & Initiative (Swim Team Captain ·

Tech Lead) · Adaptability & Resilience · Industrious Work Ethic

Languages

English Native

Spanish Proficient